

From Vertices to Edges

- Focus so far on Vertices (Nodes)
 - Actors/Agents
 - Classical "variable" sociology: Property Agent --> Behavior
 - Structural (Network) Property: Centrality, CC, Degree (Network Size), Network Composition
 - Ego Networks
 - $G = (V, E)$ with N vertices and M Edges
 - What about the edges (dyads)
 - Critical for dynamic analysis
 - tie formation and persistence (or decay)
 - flows and weighted edges based on interaction events
 - generative processes - network evolution model (rules)
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The Problem

- Binary arc data (presence/absence)
 - Dyadic Census - 3 states
 - Null Dyad
 - Asymmetric Dyad
 - Mutual Dyad
 - Tie formation (link prediction) & persistence
 - Longitudinal Data
 - Formation or Persistence
 - Next time: (choice) homophily --> tie formation
 - Prior work
 - Mutuals persist, asymmetric --> null than mutual (Hallinan1978)
 - 1997 Social Networks Special Issue (ego networks)
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Burt: Tie Decay

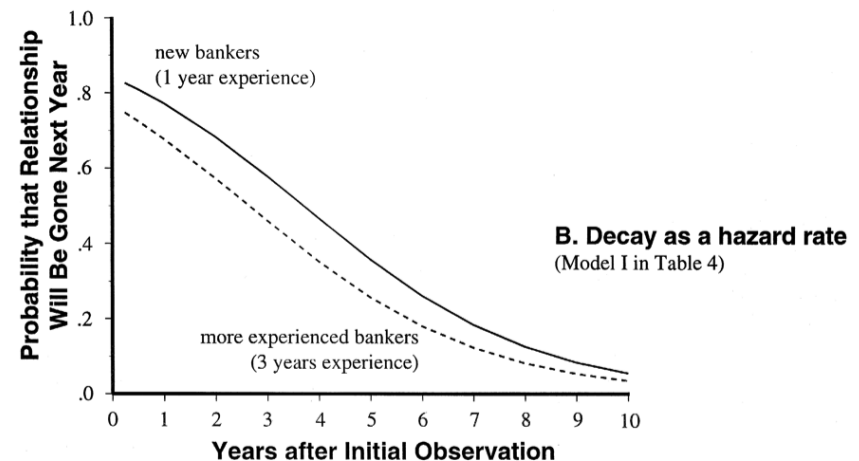
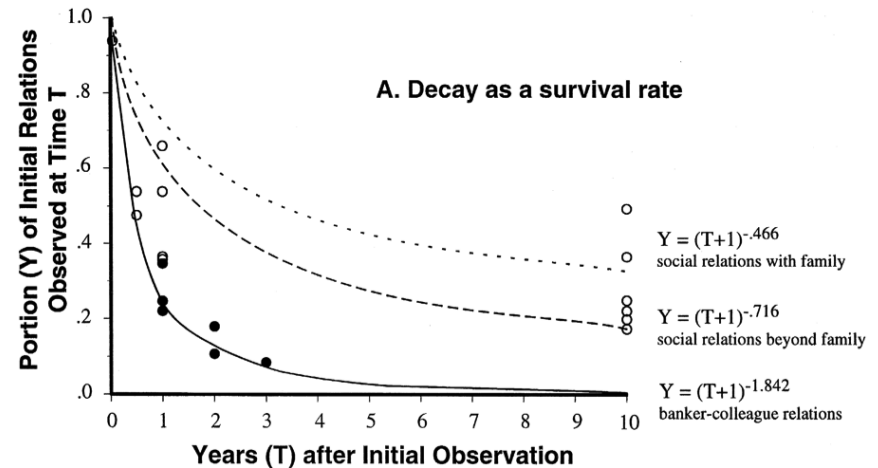
- formation vs. persistence
- liability of newness:
 - longer in a state less likely to leave the state

Table 3

Decay hazard rates

(a) This row describes colleague relationships cited in the first panel. (b) This row describes relations cited in the second panel, but not in the first panel. (c) This row describes relations cited in the third panel, but not in the second. (d) These are the relations cited this year that are at risk of not being re-cited next year. (e) These are the relations at risk that were not re-cited. (f) This is column (e) divided by (d), the proportion of relations at risk that were not re-cited.

Years observed (T)	Panel in which first observed (P)	Relationships at risk ^d	Relationships that decay ^e	Hazard rate ^f
1	1 ^a	12,655	9526	0.753
2	1 ^a	3129	2246	0.718
3	1 ^a	883	467	0.529
1	2 ^b	3835	3173	0.827
2	2 ^b	662	398	0.601
1	3 ^c	1545	865	0.560
Total		22,709	16,675	0.734



Burt: Tie Decay - Predictors

- Node Age
 - Tie Age
 - Strength (subjective measure)
 - Colleague availability
 - Homophily
 - minimal rank homophily
 - Gender: M-M lower
 - Age
 - Embeddedness (number of third parties connecting A & B)
 - lowers decay, increases persistence
 - explains homophily effects
 - disruptive embedding (cascading tie decay)
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Social Ties and Interaction Events

- What is a social tie?
 - Dyadic social interactions
 - playing tennis, having sex, talking with, collaborating together on, exchanging with, giving to, receiving from)
 - Events: instances of interaction
 - event traits: initiating, waiting times, sequencing
 - Communicative interactions
 - Initiating, who provides information, time talking, length of messages
 - Aggregating events by time period
 - Tie: at least one event in a given time period
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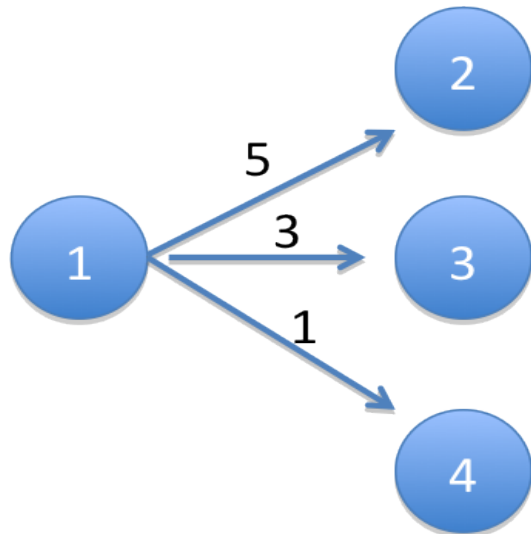
Social Ties and Interaction Events

- Tie: Link through which events occur
 - Through ties events (interactions) "flow"
 - Directed flow: initiated of event, sender and receiver
 - Strong ties: Large flows (number of events), bandwidth
 - Reciprocal ties: Balanced flows
 - Balanced counts aggregated from information on the occurrence and traits of interactional events within a time window
 - A temporal event sequence $A \rightarrow B \rightarrow A \rightarrow B$
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Directed Weighted Edge Graphs

Arc ($x_{ij} = 1$): if there was an interaction between i and j within a given time period and i initiated that interaction (e.g., a call from i to j)

Outflow Parameter (m_{ij}): Count of the number of interactions between i and j that were initiated by i in a given time period (e.g., number of calls made by i to j)



$$x_{12} = x_{13} = x_{14} = 1$$

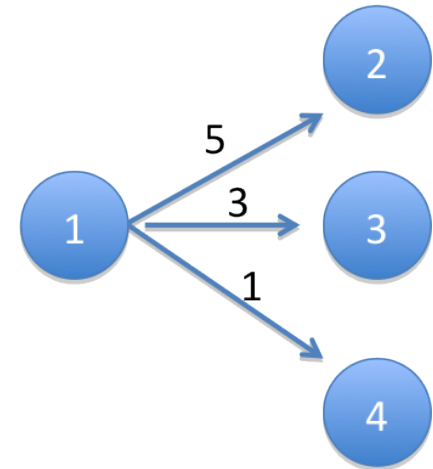
$$m_{12} = 5$$

$$m_{13} = 3$$

$$m_{14} = 1$$

Node Level Properties

- **Out-degree (d_i):** count of number of j nodes that i has initiated interaction with in a given time period (e.g., number of people i called)
- **Vertex Strength (m_{i+}):** activity level of a node, i.e. Count of the number of interactions initiated by i with all i 's j neighbors in a given time period (e.g., total number of calls made by i)



$$d_1 = 3$$

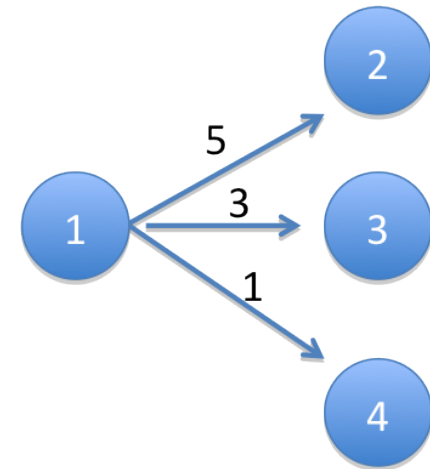
$$m_{i+} = \sum_j m_{ij}$$

$$m_{1+} = 5 + 3 + 1 = 9$$

Arc Flows & Vertex Strength

$$p_{ij} = m_{ij}/m_{i+} \quad \sum_j p_{ij} = 1$$

conditional transition probability:
probability, given that an event has
been initiated by i, that it is with j.



$$m_{12} = (m_{1+}) (p_{12})$$

$$5 = 9 \cdot (5/9)$$

$$p_{12} = 5/9 = .556$$

$$p_{13} = 3/9 = .333$$

$$p_{14} = 1/9 = .111$$

Tie (Arc) Strength

Arc Flow: $m_{ij} = (m_{i+}) (p_{ij})$

Transition Probability: p_{ij}

Relative Strength: $m_{ij}/\max(m_{ik})$

Vertex Strength: m_{i+}

Dyadic Edge Strength
(bandwidth)

$$s_{ij} = \sqrt{m_{ij}m_{ji}}$$

Reciprocity: Simple Model

Norm of Reciprocity: Agents more likely to dissolve a tie if non-reciprocal

$$R_{ij}(1) = |m_{ij} - m_{ji}|$$

$$R_{ij}(2) = \ln(m_{ij}) - \ln(m_{ji}) = \ln(m_{ij}/m_{ji})$$

Reciprocity: Behavioral Model

- Why would j initiate interaction (call) i more (less) than i initiates interaction?
- Local knowledge:
 - i knows: m_{ij} , m_{ji} , d_i , m_{i+}
 - i does not know: d_j , m_{j+}
 - assumes $m_{j+} = m_{i+}$, $d_j = d_i$
- Weight m_{ij} by m_{j+}/m_{i+}
$$R_{ij}(3) = |m_{ij}(m_{j+}/m_{i+}) - m_{ji}|$$
$$R_{ij}(4) = |\ln[m_{ij}(m_{j+}/m_{i+})] - \ln[m_{ji}]|$$
$$= |\ln[(m_{ij}/m_{i+})/(m_{ji}/m_{j+})]|$$
$$= |\ln(p_{ij}/p_{ji})| = |\ln(p_{ij}) - \ln(p_{ji})|$$

Reciprocity: Information Gain

$$R_{ij} = -p_{ij} \log_2(p_{ij}) - (1 - p_{ij}) \log_2(1 - p_{ij})$$

where
$$p_{ij} = \frac{m_{ij}}{m_{ij} + m_{ji}}$$

$$0 \geq R_{ij} \leq 1$$

$$R_{ij} = 1 \Leftrightarrow p_{ij} = .5$$

Tie Decay Revisited

Cell phone data

Table 1

Basic graph-statistics of the cell-phone network.

Statistic	Value
Average clustering coefficient ($d_i \geq 2$)	0.24
Median clustering coefficient	0.14
Average out-degree	4.2
Median out-degree	3
Average total degree	6.3
Median total degree	3
Number of vertices	4,833,408
Number of edges	16,564,958

Possible Predictors of Tie Decay

Table 2

List of features to be used in predicting edge persistence/decay $i \rightarrow j$.

Feature	Description	Range	Median
Vertex level			
d_i	Out degree of i (ego-network range)	1–49	2
d_j	Out degree of j (ego-network range)	0–49	1
c_i	Number of calls made by i (gregariousness)	1–1366	22
c_j	Number of calls made by j (gregariousness)	0–1366	22
Dyad level			
c_{ij}	Calls from i to j (directed edge strength)	1–1341	2
c_{ji}	Calls from j to i (reciprocated edge strength)	0–1341	1
p_{ij}	Proportion of i 's calls that go to j (c_{ij}/c_i)	0–1	0.15
p_{ji}	Proportion of j 's calls that go to i (c_{ji}/c_j)	0–1	0.08
Triad level			
cn	Number of common neighbors between i and j (edge embeddedness)	0–46	0
in	Number of i 's neighbors that call j (directed edge embeddedness)	0–39	1
jn	Number of j 's neighbors that call i (directed edge embeddedness)	0–39	0
$injn$	Number of calls that i 's neighbors make to j 's neighbors (2nd order embeddedness)	0–274	7
$jnin$	Number of calls that j 's neighbors make to i 's neighbors (2nd order embeddedness)	0–274	6
Temporal			
$fdate$	Normalized time of first call from i to j (edge newness)	0–1	0.26
$edate$	Time of last call from i to j (edge freshness)	0–1	0.74

Machine Learning Logic

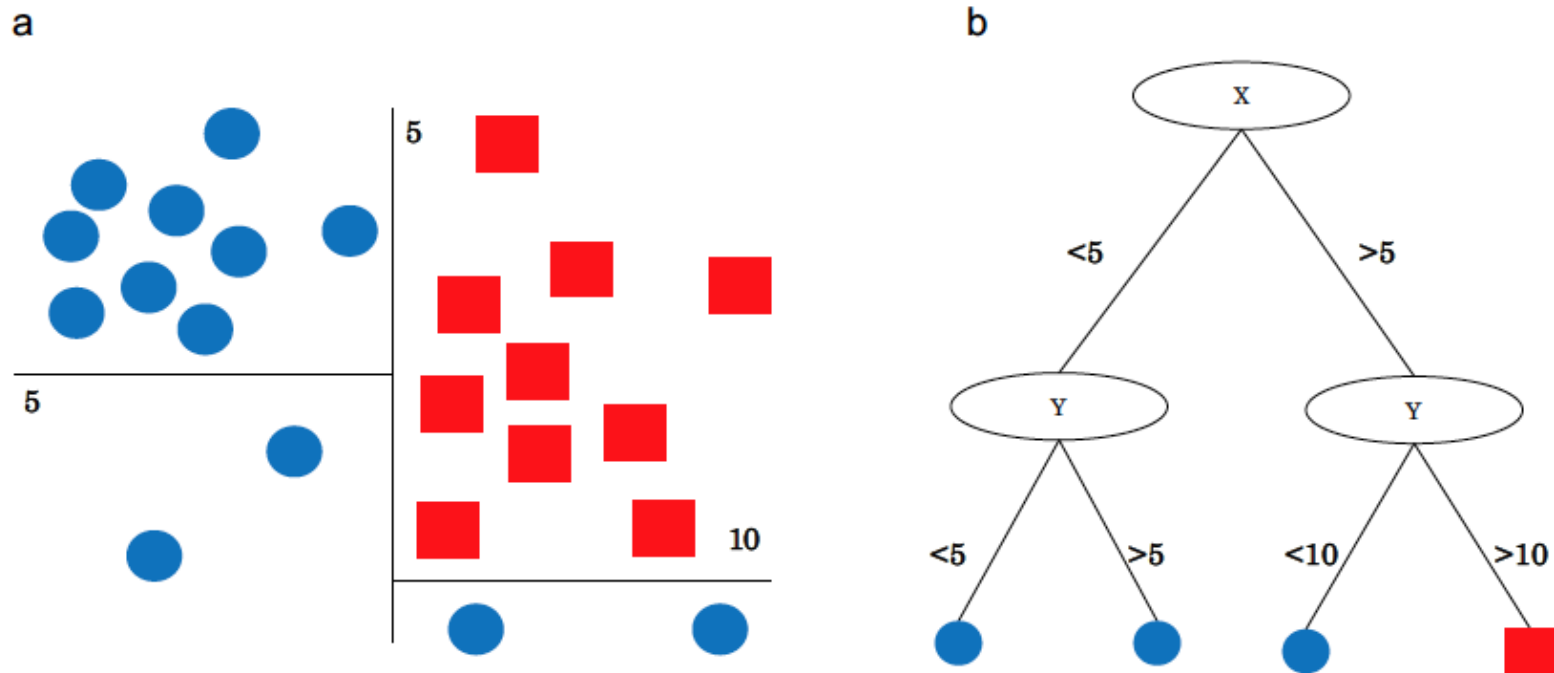


Fig. 2. (a) Toy classification dataset and (b) resulting decision tree.

Regression Results

Table 5

Logistic regression coefficients of the effect of each feature in predicting edge-persistence.

Feature	Description	β	Odds ($\exp(\beta)$)
d_i	Degree of i	-0.0335	0.9671
d_j	Degree of j	0.0057	1.0057
c_i	Calls made by i	0.0003	1.0003
c_j	Calls made by j	-0.0013	0.9987
c_{ij}	Calls from i to j	0.0373	1.0380
c_{ji}	Calls from j to i	0.0229	1.0232
p_{ij}	Proportion of i 's calls that go to j	0.0504	1.05178
p_{ji}	Proportion of j 's calls that go to i	0.8521	2.3446
in	Number of i 's neighbors that call j	0.1409	1.1513
jn	Number of j 's neighbors that call i 's neighbors	0.0877	1.0917
cn	Number of common neighbors between i and j	0.0525	1.05391
$jnin$	Number of j 's neighbors that call i 's neighbors	-0.0366	0.9641
$injn$	Number of i 's neighbors that call j 's neighbors	0.0416	1.0425
$fdate$	Time of first call from i to j	-2.3021	0.1000
$edate$	Time of last call from i to j	2.9218	18.5747

Decision-Tree Results

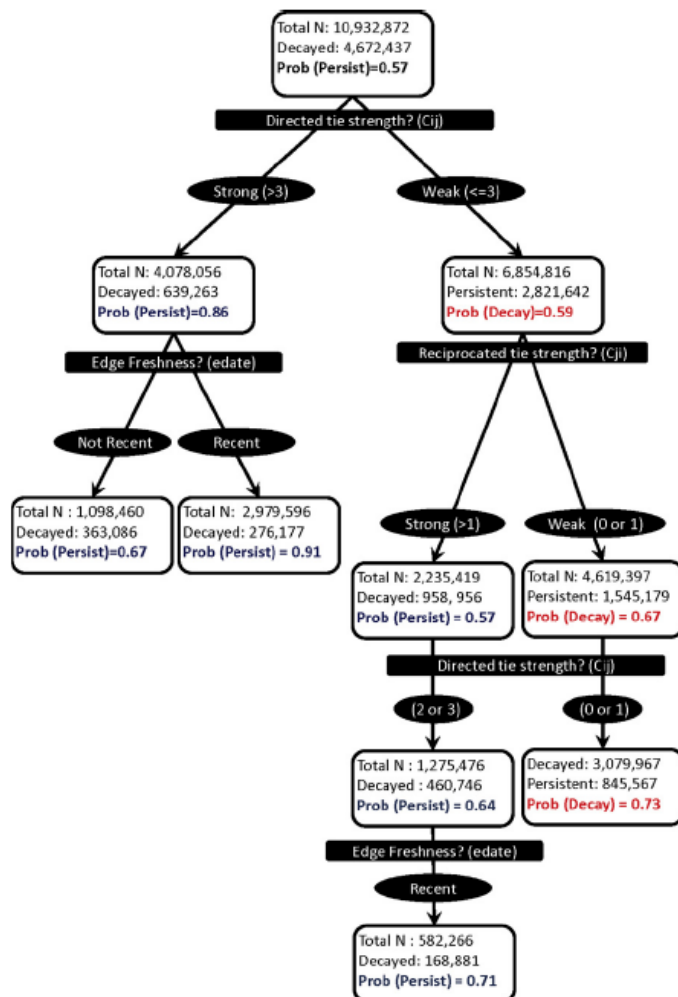
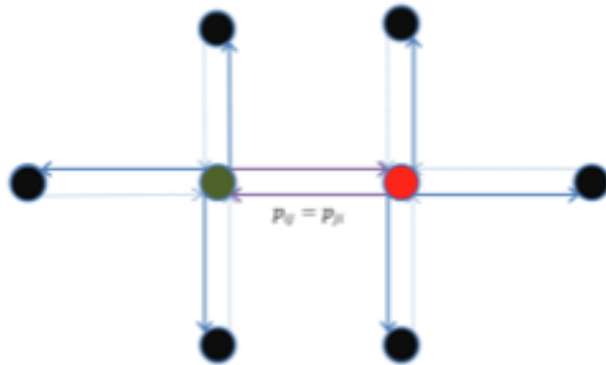


Fig. 5. Selected leaves of the best-fitting decision-tree obtained from the training set.

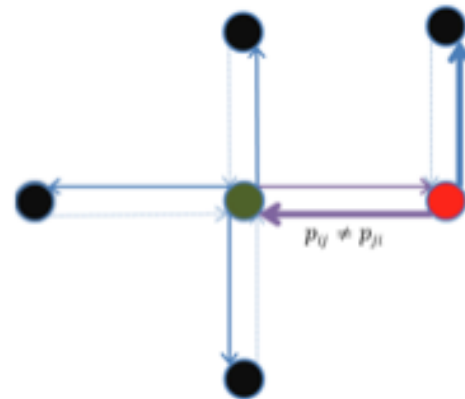
- Path 1
 - directed tie strength > 3
 - recent activation
- Path 2
 - somewhat weak tie
 - reciprocated
 - recent activation
- Strong ties persist
- Weaker ties can persist if they become reciprocate
 - They then can also strengthen with time
- Net of this, embeddedness not important.

Explaining (Non-)Reciprocity

Out-Degree Mismatch



Reciprocity as a Result of Out-Degree Matching



Non Reciprocity as a Result of Out-Degree Mismatch

Explaining (Non-)Reciprocity

Unequal dispersion: $m_{ij} \neq m_{ik}$

$$R_{ij} = \left| \ln \left[\frac{w_{ij}}{w_{ji}} \frac{w_{j+}}{w_{i+}} \right] \right|$$

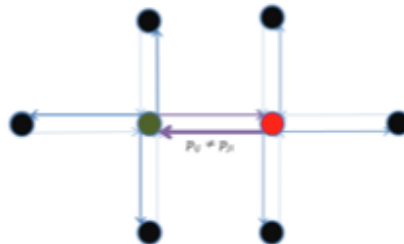
Ratio of w_{ij} to w_{ji} adjusted by ratio of j to i 's activity level

$$\hat{w}_{ij} = \frac{w_{i+}}{k_i^{out}} \quad \hat{p}_{ij} = \frac{1}{k_i^{out}}$$

$$\hat{R}_{ij} = \left| \ln(k_j^{out}) - \ln(k_i^{out}) \right|$$

when equidispersion, vertex strength drops out and reciprocity a function of degree differences

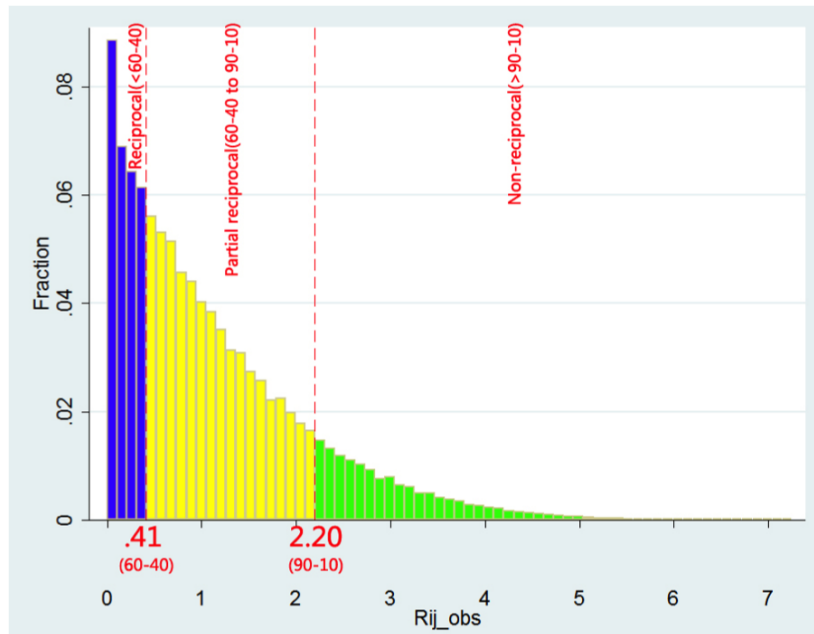
- degree differences --> non-reciprocity
- unequal dispersion --> non-reciprocity



Non-reciprocity as a result of directed weight-mismatch

- same degree
- red vertice overdisperses

Reciprocity Distribution



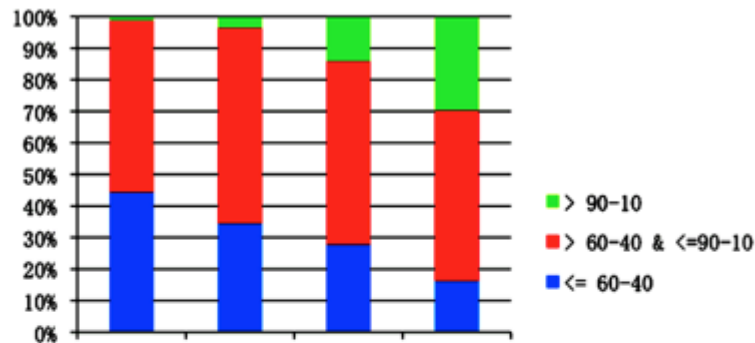
- 27% of mutual dyads are reciprocal
- 58% partially reciprocal
- 14% non-reciprocal

Four Variants of Reciprocity

	Assortativity	Non-Assortativity
<u>Equidispersion</u>	$\hat{R}_{ij}^{obs}$	$\hat{R}_{ij}^{rw}$
<u>Non-equidispersion</u>	R_{ij}^{obs}	R_{ij}^{rw}

1. Observed Network: Assortative & non-equidispersed
 2. Equi-dispersed Model
 3. Rewired: Non-assortative and non-equidisperses
 4. Rewired with Equi-dispersion
-

Results



	$\hat{R}^{obs}$	$\hat{R}^{rw}$	R^{obs}	R^{rw}
Dispersion	=	=	≠	≠
Assortativity	Yes	No	Yes	No

$$\hat{R}_{ij}^{obs} < \hat{R}_{ij}^{rw} \leq R_{ij}^{obs} < R_{ij}^{rw}$$

- Social Networks exhibit non-reciprocity because
 - not fully degree assortative
 - Unequal dispersion

Predictiveness

Table 3

Information gain of each feature for predicting short-term edge decay using four weeks of data. The information gain measures the conditional ability of that feature to predict edged decay in the subsequent week within levels of the other features.

Feature	Description	Info. gain
d_i	Degree of i .	0.00235
d_j	Degree of j .	0.00234
c_i	Calls made by i .	0.01181
c_j	Calls made by j .	0.00449
c_{ij}	Calls from i to j .	0.17948
c_{ji}	Calls from j to i .	0.12823
p_{ij}	Proportion of i 's calls that go to j	0.13318
p_{ji}	Proportion of j 's calls that go to i .	0.12043
in	Number of i 's neighbors that call j .	0.02478
jn	Number of j 's neighbors that call i 's neighbors.	0.02303
cn	Number of common neighbors between i and j .	0.02441
$jnin$	Number of j 's neighbors that call i 's neighbors.	0.01501
$injn$	Number of i 's neighbors that call j 's neighbors.	0.00493
$fdate$	Time of first call from i to j .	0.05104
$edate$	Time of last call from i to j .	0.09954